

WEB PROGRAMMING ASSIGNMENT

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CODE 1-

```
import React, { useState } from "react";

export default function App() {
  const [products, setProducts] = useState([]);
  const [form, setForm] = useState({
    id: "",
    name: "",
    category: "",
    price: "",
    stock: ""
  });

  const handleChange = (e) => {
    setForm({ ...form, [e.target.name]: e.target.value });
  };

  const addProduct = () => {
    if (!form.id || !form.name) return alert("Fill all fields");

    if (products.some(p => p.id === form.id)) {
      return alert("Duplicate ID not allowed");
    }

    setProducts([...products, form]);
    setForm({ id: "", name: "", category: "", price: "", stock: "" });
  };
}
```

```

};

const deleteProduct = (id) => {
  setProducts(products.filter(p => p.id !== id));
};

return (
  <div>
    <h2>Inventory</h2>

    <input name="id" placeholder="ID" value={form.id} onChange={handleChange} />
    <input name="name" placeholder="Name" value={form.name} onChange={handleChange} />
    <input name="category" placeholder="Category" value={form.category}
    onChange={handleChange} />
    <input name="price" placeholder="Price" value={form.price} onChange={handleChange} />
    <input name="stock" placeholder="Stock" value={form.stock} onChange={handleChange} />

    <button onClick={addProduct}>Add</button>

    {products.map(p => (
      <div key={p.id} style={{ color: p.stock < 5 ? "red" : "black" }}>
        {p.name} - {p.stock}
        <button onClick={() => deleteProduct(p.id)}>Delete</button>
      </div>
    ))}
  </div>
);
}

```

CODE 2-import React, { useState } from "react";

```

export default function App() {
  const [students, setStudents] = useState([]);
  const [name, setName] = useState("");
  const [marks, setMarks] = useState(["", "", "", "", ""]);
  const [filter, setFilter] = useState("");

  const addStudent = () => {
    const nums = marks.map(Number);
    const total = nums.reduce((a, b) => a + b, 0);
    const avg = total / 5;
    const grade = avg > 80 ? "A" : avg > 50 ? "B" : "C";

    setStudents([...students, { name, nums, total, avg, grade }]);
  };

  const filtered = students.filter(s => s.avg >= filter);

  return (
    <div>
      <h2>Student Analyzer</h2>

      <input placeholder="Name" onChange={e => setName(e.target.value)} />
      {marks.map((m, i) => (
        <input key={i} onChange={e => {
          const newMarks = [...marks];
          newMarks[i] = e.target.value;
          setMarks(newMarks);
        }} />
      ))}

      <button onClick={addStudent}>Add</button>
    </div>
  );
}

```

```

<input placeholder="Filter Avg >" onChange={e => setFilter(e.target.value)} />

{filtered.map((s, i) => (
  <div key={i}>
    {s.name} - Avg: {s.avg} - {s.grade} - {s.avg >= 40 ? "Pass" : "Fail"}
  </div>
))}
</div>
);
}

```

CODE 3-import React, { useState, useRef, useEffect } from "react";

```

export default function App() {
  const [messages, setMessages] = useState([]);
  const [text, setText] = useState("");
  const chatEndRef = useRef();

  const sendMessage = () => {
    const msg = {
      text,
      time: new Date().toLocaleTimeString(),
      sender: "me"
    };
    setMessages([...messages, msg]);
    setText("");
  };

  useEffect(() => {

```

```

chatEndRef.current?.scrollIntoView({ behavior: "smooth" });
}, [messages]);

return (
  <div style={{ height: "300px", overflowY: "scroll" }}>
    {messages.map((m, i) => (
      <div key={i} style={{ textAlign: m.sender === "me" ? "right" : "left" }}>
        {m.text} {m.time}
      </div>
    ))}
    <div ref={chatEndRef}></div>

    <input value={text} onChange={e => setText(e.target.value)} />
    <button onClick={sendMessage}>Send</button>
  </div>
);
}

```

CODE 4-

```

import React, { useState } from "react";

export default function App() {
  const [step, setStep] = useState(1);
  const [data, setData] = useState({
    name: "",
    email: "",
    address: ""
  });
}

```

```
const next = () => {  
  if (step === 1 && !data.name) return alert("Enter name");  
  setStep(step + 1);  
};
```

```
const back = () => setStep(step - 1);
```

```
return (  
  <div>  
    {step === 1 && (  
      <>  
        <input placeholder="Name" onChange={e => setData({ ...data, name: e.target.value })} />  
        <button onClick={next}>Next</button>  
      </>  
    )}  
  
    {step === 2 && (  
      <>  
        <input placeholder="Address" onChange={e => setData({ ...data, address: e.target.value })} />  
        <button onClick={back}>Back</button>  
        <button onClick={next}>Next</button>  
      </>  
    )}
```

```
    {step === 3 && (  
      <>  
        <h3>Review</h3>  
        <p>{data.name}</p>  
        <p>{data.address}</p>  
        <button onClick={back}>Back</button>  
        <button onClick={() => alert("Submitted")}>Submit</button>  
      </>  
    )}
```

```

        </>
    )}
</div>
);
}

```

CODE 5-

```

import React, { useState } from "react";

export default function App() {
  const [tasks, setTasks] = useState([]);
  const [text, setText] = useState("");
  const [priority, setPriority] = useState("Low");

  const addTask = () => {
    setTasks([...tasks, { text, priority, done: false }]);
  };

  const toggleDone = (i) => {
    const updated = [...tasks];
    updated[i].done = !updated[i].done;
    setTasks(updated);
  };

  const grouped = {
    High: tasks.filter(t => t.priority === "High"),
    Medium: tasks.filter(t => t.priority === "Medium"),
    Low: tasks.filter(t => t.priority === "Low")
  };

  return (

```

```
<div>

  <input onChange={e => setText(e.target.value)} />

  <select onChange={e => setPriority(e.target.value)}>
    <option>High</option>
    <option>Medium</option>
    <option>Low</option>
  </select>
```

```
<button onClick={addTask}>Add</button>
```

```
{Object.keys(grouped).map(p => (
  <div key={p}>
    <h3>{p}</h3>
    {grouped[p].map((t, i) => (
      <div key={i} onClick={() => toggleDone(i)}>
        {t.text} {t.done ? "✓" : ""}
      </div>
    ))}
  </div>
))}
</div>
);
}
```